



MATHEMATICS (EXTENSION 1)

2013 HSC Course Assessment Task 4

Friday August 16, 2013

General instructions

- Working time – 50 min.
(plus 5 minutes reading time)
- Commence each new question on a new page.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 12M3A – Mr Lam
- ☐ 12M3B – Mr Berry
- ☐ 12M3C – Mr Lin
- ☐ 12M4A – Mr Choy
- ☐ 12M4B – Mr Weiss
- ☐ 12M4C – Ms Ziazaris

STUDENT NUMBER # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	4 (ab)	4 (cd)	Total	%
MARKS	$\overline{4}$	$\overline{8}$	$\overline{6}$	$\overline{4}$	$\overline{5}$	$\overline{27}$	

Question 1 (4 Marks)	Commence a NEW page.	Marks
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In the expansion of $\left(\frac{2}{x} - \frac{x^2}{3}\right)^{12}$

- | | | |
|-----|--|----------|
| (a) | Write the expansion in sigma notation. | 2 |
| (b) | Find the term independent of x in the expansion. | 2 |

Question 2 (8 Marks)	Commence a NEW page.	Marks
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Mr Cash is borrowing \$500 000 to purchase a property. The bank will be lending him the amount based on a 6% p.a. interest rate on the balance owing, and loan term of 30 years with equal monthly repayments of \$ M . \$ A_n is the amount owing after the n -th monthly repayment.

- | | | |
|-----|---|----------|
| (a) | Show that $A_n = 500\,000 \times 1.005^n - 200M(1.005^n - 1)$. | 3 |
| (b) | Hence or otherwise, show that the repayment amount $M = \$2997.75$. | 1 |
| (c) | Show that, if Mr Cash decides to make repayments of \$3 200 on a monthly basis, that he would be able to save approximately 55 months off the term of the loan. | 2 |
| (d) | Hence or otherwise, find the amount of interest saved by making repayments of \$3 200 instead of \$2 997.75 per month. | 2 |

(For simplicity, you may assume the final repayment for the additional repayment schedule is also \$3 200).

Question 3 (6 Marks)	Commence a NEW page.	Marks
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A particle is moving on a horizontal line with velocity $v \text{ ms}^{-1}$, with v^2 given as

$$v^2 = 64 + 24x - 4x^2$$

- | | | |
|-----|--|----------|
| (a) | Prove that the particle is moving in simple harmonic motion. | 2 |
| (b) | Find the centre of the motion. | 1 |
| (c) | Find the period and amplitude. | 3 |

Question 4 (9 Marks)

Commence a NEW page.

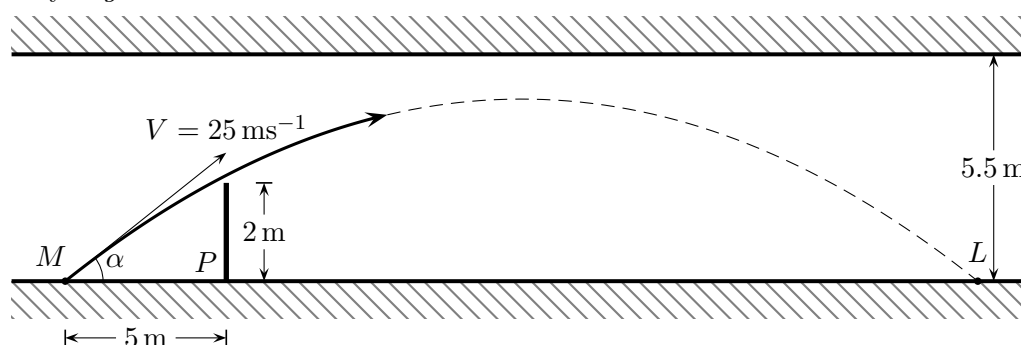
Marks

A movie scene is to be filmed inside a tunnel of height 5.5 m. In this scene, a stunt man (of negligible height) will ride a motorcycle off an inclined plane (in the shape of a wedge, of negligible height) positioned at point M at α degrees, with velocity $V = 25 \text{ ms}^{-1}$.

The stunt man must just clear a policeman of height 2 m, standing 5 m away from the inclined plane at point P .

The stunt man will then land safely at another point L in the tunnel.

Assume that there is no air resistance in the tunnel, and that the acceleration due to gravity is $g = 10 \text{ ms}^{-2}$.



You may also assume the equations of motion are $\begin{cases} x = 25t \cos \alpha \\ y = 25t \sin \alpha - 5t^2 \end{cases}$

and the equations for the velocity are $\begin{cases} \dot{x} = 25 \cos \alpha \\ \dot{y} = 25 \sin \alpha - 10t \end{cases}$.

- (a) Show that the trajectory can be expressed as **2**

$$y = -\frac{x^2}{125} \sec^2 \alpha + x \tan \alpha$$

- (b) Hence show that $\tan^2 \alpha - 25 \tan \alpha + 11 = 0$. **2**

- (c) Hence or otherwise, show that the value(s) of α (correct to the nearest degree) are **2**

$$\alpha = 24^\circ \text{ or } 88^\circ$$

- (d) Explain (with mathematical reasoning) which of these values of α is valid. **3**

For simplicity, the rounded off values from part (c) may be used in your explanation

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Suggested Solutions

Question 1 (Ziaziaris)

(a) (2 marks)

$$\begin{aligned} & \left(\frac{2}{x} - \frac{x^2}{3} \right)^{12} \\ &= \sum_{k=0}^{12} \binom{12}{k} 2^k (x^{-1})^k (-1)^{12-k} (3^{-1})^{12-k} (x^2)^{12-k} \\ &= \sum_{k=0}^{12} \binom{12}{k} 2^k (-1)^{12-k} 3^{k-12} x^{-k+24-2k} \\ &= \sum_{k=0}^{12} \binom{12}{k} 2^k (-1)^{12-k} 3^{k-12} x^{24-3k} \end{aligned}$$

(b) (2 marks) Term independent of x appears when

$$\begin{aligned} 24 - 3k &= 0 \\ 3k &= 24 \\ k &= 8 \end{aligned}$$

Hence, the constant term is

$$\binom{12}{8} 2^8 3^{-4}$$

Question 2 (Lin)

(a) (3 marks)

- At the end of the first month:
 $r = \frac{0.06}{12} = 0.005$ p.m.

$$A_1 = 500\,000 \times 1.005 - M$$

- At the end of the second month:

$$\begin{aligned} A_2 &= A_1 \times 1.005 - M \\ &= (500\,000 \times 1.005 - M) \times 1.005 - M \\ &= 500\,000 \times 1.005^2 - M(1 + 1.005) \end{aligned}$$

- At the end of the third month:

$$\begin{aligned} A_3 &= A_2 \times 1.005 - M \\ &= 500\,000 \times 1.005^3 \\ &\quad - M(1 + 1.005 + 1.005^2) \end{aligned}$$

- At the end of n months,

$$A_n = 500\,000 \times 1.005^n - M \underbrace{(1 + 1.005 + 1.005^2 + \cdots + 1.005^{n-1})}_{S_n: r = 1.005, a = 1, n \text{ terms}}$$

$$\begin{aligned} S_n &= \frac{a(r^n - 1)}{r - 1} \\ &= \frac{1.005^n - 1}{1.005 - 1} \\ &= 200(1.005^n - 1) \end{aligned}$$

$$\therefore A_n = 500\,000 \times 1.005^n - 200M(1.005^n - 1)$$

(b) (1 mark)

At the end of the loan of 30 years, $n = 360$,
 $A_{360} = 0$:

$$\begin{aligned} 500\,000 \times 1.005^{360} - 200M(1.005^{360} - 1) &= 0 \\ 200M(1.005^{360} - 1) &= 500\,000 \times 1.005^{360} \\ \therefore M &= \frac{500\,000 \times 1.005^{360}}{200(1.005^{360} - 1)} = \$2997.75 \end{aligned}$$

(c) (2 marks) If $M = \$3\,200$,

$$A_n = 500\,000 \times 1.005^n - 200(3\,200)(1.005^n - 1)$$

At the end of the loan, $A_n = 0$:

$$\begin{aligned} 500\,000 \times 1.005^n - 200(3\,200)(1.005^n - 1) &= 0 \\ 1.005^n(500\,000 - 640\,000) + 640\,000 &= 0 \\ \therefore 1.005^n \times 140\,000 &= 640\,000 \end{aligned}$$

$$1.005^n = \frac{64}{14}$$

$$n \log 1.005 = \log \frac{64}{14}$$

$$n = \frac{\log \frac{32}{7}}{\log 1.005} = 304.72 \approx 305 \text{ months}$$

Previous repayment schedule had 360 months. Increasing repayments to \$3 200 per month now only takes 305 months. Hence a time saving of 55 months.

(d) (2 marks)

- \$2 997.75 per month repayments:
 - Total repaid:

$$\$2997.75 \times 360 = \$1\,079\,190$$

- Total interest:

$$\$1\,079\,190 - \$500\,000 = \$579\,190$$

- \$3 200 per month repayments:
 - Total repaid:

$$\$3200 \times 304.72 = \$975\,104$$

Students may use 305 months:
\$976 000

- Total interest:

$$\$975\,104 - \$500\,000 = \$475\,104$$

(\$476 000)

- Total interest saved over 55 months:

$$\$579\,190 - \$475\,104 = \$104\,086$$

(\$103 190 if 305 mths used)

Question 3 (Berry)

- (a) (2 marks)

$$\begin{aligned}\ddot{x} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\ &= \frac{d}{dx} \left(\frac{1}{2} (64 + 24x - 4x^2) \right) \\ &= \frac{d}{dx} (32 + 12x - 2x^2) \\ &= 12 - 4x \\ &= -4(x - 3) \\ &\equiv -n^2(x - k)\end{aligned}$$

As acceleration is proportional to, and directed against the motion, the particle is moving in simple harmonic motion.

- (b) (1 mark) Centre: $x = 3$.

- (c) (3 marks)

$$\begin{aligned}v^2 &= 64 + 24x - 4x^2 \\ &= -4(x^2 - 6x - 16) \\ &= -4(x^2 - 6x + 9 - 25) \\ &= -4((x - 3)^2 - 25) \\ &= 4(25 - (x - 3)^2) \\ &\equiv n^2(a^2 - (x - k)^2) \\ T &= \frac{2\pi}{n} = \frac{2\pi}{2} = \pi \\ a &= 5\end{aligned}$$

Question 4 (Weiss (a)(b), Choy (c)(d))

- (a) (2 marks)

$$\begin{cases} x = 25t \cos \alpha & (1) \\ y = 25t \sin \alpha - 5t^2 & (2) \end{cases}$$

Change subject of (1) to t , and substitute to (2):

$$\begin{aligned}t &= \frac{x}{25 \cos \alpha} \\ y &= 25 \left(\frac{x \sin \alpha}{25 \cos \alpha} \right) - 5 \left(\frac{x}{25 \cos \alpha} \right)^2 \\ &= x \tan \alpha - \frac{5x^2}{625 \cos^2 \alpha} \\ &= x \tan \alpha - \frac{x^2}{125} \sec^2 \alpha\end{aligned}$$

- (b) (2 marks) When $x = 5$, $y = 2$:

$$\begin{aligned}2 &= 5 \tan \alpha - \frac{5^2}{125} (1 + \tan^2 \alpha) \\ \frac{2}{\times 5} &= 5 \tan \alpha - \frac{1}{5} (1 + \tan^2 \alpha) \\ &\quad \times 5 \\ 10 &= 25 \tan \alpha - 1 - \tan^2 \alpha \\ \therefore \tan^2 \alpha - 25 \tan \alpha + 11 &= 0\end{aligned}$$

- (c) (2 marks) Let $z = \tan \alpha$.

$$\begin{aligned}z^2 - 25z + 11 &= 0 \\ z &= \frac{25 \pm \sqrt{25^2 - 4(1)(11)}}{2} \\ &= \frac{25 \pm \sqrt{581}}{2} \\ &= \frac{25 \pm 24.1 \dots}{2}\end{aligned}$$

$$\tan \alpha = 24.5519 \text{ or } 0.4480$$

$$\begin{aligned}\therefore \alpha &= 87.6676^\circ \text{ or } 24.1337^\circ \\ &\approx 88^\circ \text{ or } 24^\circ\end{aligned}$$

- (d) (3 marks)

- If $\alpha = 88^\circ$, check maximum height – i.e $\dot{y} = 0$

$$\begin{aligned}\dot{y} &= 0 = 25 \sin 88^\circ - 10t \\ \therefore 10t &= 25 \sin 88^\circ \\ t &= \frac{25 \sin 88^\circ}{10} \approx 2.498 \text{ s}\end{aligned}$$

Check maximum height attainable with this angle of inclination:

$$\begin{aligned} y &= 25t \sin 88^\circ - 5t^2 \\ &= 25(2.498) \sin 88^\circ - 5(2.498)^2 \\ &\approx 31.2 \text{ m} \end{aligned}$$

i.e. if the inclined plane is at 88° , the stunt man will hit the roof well before he reaches his maximum height. Therefore an unsafe landing.

- If $\alpha = 24^\circ$, check maximum height – i.e. $\dot{y} = 0$

$$\begin{aligned} \dot{y} &= 0 = 25 \sin 24^\circ - 10t \\ \therefore 10t &= 25 \sin 24^\circ \\ t &= \frac{25 \sin 24^\circ}{10} \approx 1.0168 \text{ s} \end{aligned}$$

Check maximum height attainable with this angle of inclination:

$$\begin{aligned} y &= 25t \sin 24^\circ - 5t^2 \\ &= 25(1.0168) \sin 24^\circ - 5(1.0168)^2 \\ &\approx 5.1698 \text{ m} \end{aligned}$$

The maximum height attained by the stunt man when $\alpha = 24^\circ$ will be approx 5.2 m, well below the 5.5 m roof. Therefore he will land safely inside the tunnel.